Ashish Kumar

Contact Information

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Education

PhD Economics, Johns Hopkins University, (2022-)
Advisor: Christopher D. Carroll.

MA Economics, Ashoka University, 2020

GPA: 3.7/4 (magna cum laude)

- Dissertation Topic - Production Linkages & Volatility in an Economy:A Network based Exploration Link, Advisor : Prof. Bhaskar Dutta

BA Economics, Punjabi University, 2017
Percentage: 82% (First class with distinction)

Experience

Summer Intern, Fund Internship Program (FIP), International Monetary Fund, June 2026-August 2026

Research question: How do differing demographic trends across countries shape the equilibrium quantity and price of safe assets globally — and, in turn, the level and dynamics of the global equity premium.

Workshop Participation: Heterogeneous-Agent Macroeconomics using SSJ Toolkit (Frankfurt 2024)Link

Research Assistant to Prof. Andrew Ching, July 2023-August 2023

Research Assistant to Prof Christopher Carroll, December 2022-January 2023

Research Assistant, Indian Health Outcomes, Public Health & Economics Research Centre, June 2021-May 2022

Research Assistant, Misra Centre for Financial Markets and Economy, IIM-A, April 2021-May 2021

Research Assistant, Centre for Management of Health Services, IIM-A, May 2020-March 2021

Research Intern, ICTS-TIFR, Bengaluru, May-July 2019 Link

Teaching Experience

Macroeconomic Theory (Undergraduate), TA to Laurence Ball (JHU)- Spring 2026

Macroeconomic Theory I (Ph.D.), TA to Christopher D. Carroll (JHU)- Fall 2025

Elements of Macroeconomics, TA to Hellen Seshie-Nasser(JHU)- Spring 2025

Macroeconomic Theory I (Ph.D.) ,TA to Christopher D. Carroll (JHU)- Fall 2024

Technical Skills

Julia, Python, R, Fortran

Publications

A. Kumar, A. S. Chakrabarti, A. Chakraborti, and T. Nandi, Distress propagation on production networks: Coarse-graining and modularity of linkages, Physica A, (2021) 568. (Link)

Working Papers

The Macroeconomic Consequences of Structural Change: Pension Reforms, Intergenerational Transfers, and Wealth Inequality (JMP) (*Work in Progress*)

The United States has undergone three intertwined structural transformations since the 1970s: a dramatic shift from defined benefit (DB) to defined contribution (DC) private pensions, a rise in labor income inequality, and a surge in intergenerational wealth transfers. This paper asks how these forces — individually and in combination — shaped macroeconomic aggregates over the transition, including the interest rate, the equity premium, and the distribution of wealth. I build a heterogeneous-agent, general equilibrium overlapping-generations model featuring portfolio choice between liquid and illiquid assets, endogenous bequests, and three pension regimes (DC, DB, and no pension). The model is calibrated to the U.S. economy and used to simulate the perfect-foresight transition path from a DB-dominated steady state to a DC-dominated one. Three channels drive the aggregate dynamics: the shift from mandatory to voluntary retirement saving, which interacts with progressive social insurance to widen wealth dispersion; endogenous rate-of-return heterogeneity arising from portfolio choice inside DC accounts; and the unlocking of retirement balances as bequestable wealth. By tracing how equilibrium prices respond along the transition path, the paper quantifies the contribution of pension reform — relative to rising income inequality and intergenerational transfers — to observed trends in capital accumulation, asset returns, and wealth concentration.

Tax Reforms and Return Heterogeneity: Quantitative Insights on Wealth Inequality and Welfare (GitHub)

This paper extends the analysis of wealth taxation reforms by Guvenen et al. (2023) by relaxing the assumptions imposed on non-homothetic preferences and examining the implications of bequest taxes. Using a quantitative general equilibrium overlapping generations model with persistent rate of return heterogeneity, we replace capital income taxation with wealth and inheritance taxes in two separate reforms to examine their efficiency and distributional impacts. The model shows that wealth taxation, compared to capital income taxation, enhances aggregate consumption, capital stock, and overall welfare, especially benefiting younger individuals with high investment abilities. In contrast, inheritance taxes yield mixed results, with modest welfare improvements but a reduction in aggregate capital and output. These findings contribute to the debate on optimal taxation policies by highlighting critical trade-offs associated with different tax structures.

Aging, Inequality, and Fiscal Sustainability: Labor Supply, Saving, and the Long-Run Debt Path (*Work in Progress*)

Dynamic Life-Cycle Model Replication in Julia (De Nardi, 2004) Link to Github repo
Link to webpage

Replicated De Nardi (2004), Wealth Inequality and Intergenerational Links, by developing a robust Julia implementation that solves a complex dynamic life-cycle model of consumption-saving with voluntary and accidental bequests. Verified the original results and enhanced computational efficiency, demonstrating proficiency in quantitative methods and modern programming tools.

Past Research

A Face-Off between Markets and Institutions: Evidence from En-Masse Pharmaceutical Product Ban in India (with Anindya S. Chakrabarti and Chirantan Chatterjee) (Link)

Cross-Border Spillovers from Pharmaceutical Policy in the United States: Evidence from Large Molecule Drugs in India (with Chirantan Chatterjee, Anindya S Chakraborti, and Dr. Raja Naraynan) (Link)

Exploiting the passage of the Biologics Price Competition and Innovation (BPCI) Act in 2008 as a quasi-natural experiment, this paper documents the shift in the market structure of biologic drugs in India. The BPCIA's intent was to foster competition by creating an abbreviated pathway for the entry of biosimilars into the biologics market — an incentive that reached firms in India, a hub of the biopharmaceutical industry. We show that the incentive channel created by the BPCIA led to market expansion, with new firms entering the market and greater product variety, which in turn brought down prices and reduced market concentration.

Administrative Duties

Website Administrator, Department of Economics, Johns Hopkins University, 2024-2025
Managed and maintained the Department of Economics website.

Co-Organizer, Macro Brown Bag Seminar, Johns Hopkins University (with Haruki Shibuya), 2024-2025

Awards and Accomplishment

Featured in Dean's List and awarded 'Excellence in Academic Achievement', Ashoka University

Batch Topper: MA Economics, Ashoka University, 2020; B.A. Economics, Punjabi University, 2017